

Simulation data generation scenario 1

Yunshan Duan

2024-03-14

Load library

```
library(ggplot2); theme_set(theme_bw())
library(dplyr)
library(Seurat)
library(scDesign2)
library(ggpubr)

data_dir <- "../data/simulation_sc1"

# colors
pal <- c("#ffb6db", "#33A02C", "#b66dff", "#FEC44F", "#41B6C4", "#8E0152", "#0868AC", "#807DBA", "#E7298A",
         "#00441B", "#525252", "#4D9221", "#8B5742", "#D8DAEB", "#7cdd2d", "#980043", "#8C96C6", "#EC70C2",
         "#FDAE61", "#1D91C0", "#A6DBA0", "#4292C6", "#BF812D", "#01665E", "#41AB5D", "#FE9929", "#252525")
names(pal) <- 1:30
```

Preprocess PBMC data set

```
if (file.exists("../pbmc/pbmc3k_final.rds")) {
  pbmc <- readRDS("../pbmc/pbmc3k_final.rds")
} else {
  cat("***Please first run 'pbmc3k.Rmd' for pbmc data preprocessing.***")
}
```

Simulation single cell data given proportions of true cell types

```
pbmc <- RunPCA(pbmc, npcs = 10)

# count matrix
count <- as.matrix(pbmc@assays$RNA@counts)
dim(count)

## [1] 13714 2638

# [1] 13714 2638

# feature matrix after PCA (dim reduction)
gene <- pbmc@reductions$pca@cell.embeddings
dim(gene)

## [1] 2638 10
```

```

# [1] 2638 10

# umap embeddings
gene_umap <- pbmc@reductions$umap@cell.embeddings
dim(gene_umap)

## [1] 2638 2

# [1] 2638 2

# cell names
cell_names <- colnames(count)

new.cluster.ids <- c("Naive CD4 T", "CD14+ Mono", "Memory CD4 T", "B", "CD8 T", "FCGR3A+ Mono",
                    "NK", "DC", "Platelet")

cell_types <- pbmc@meta.data$seurat_clusters
cell_types_char <- new.cluster.ids[cell_types]

cell_selected <- cell_names[which(cell_types_char %in% c("Naive CD4 T", "Memory CD4 T", "CD8 T", "NK") &
                                gene_umap[,1] > -5 & gene_umap[,2] < 4.8)]
length(cell_selected)

## [1] 1602

# [1] 1601

pbmc_sel <- pbmc[,cell_selected]

#####
# count matrix
count <- as.matrix(pbmc_sel@assays$RNA@counts)
dim(count)

## [1] 13714 1602

# [1] 13714 1601

# feature matrix after PCA (dim reduction)
gene <- pbmc_sel@reductions$pca@cell.embeddings
dim(gene)

## [1] 1602 10

# [1] 1601 10

# umap embeddings
gene_umap <- pbmc_sel@reductions$umap@cell.embeddings
dim(gene_umap)

## [1] 1602 2

# [1] 1601 2

# cell names
cell_names <- colnames(count)

```

```
cell_types <- pbmc_sel@meta.data$seurat_clusters
cell_types_str <- new.cluster.ids[cell_types]
```

Simulation conditions for single cell data and save the simulated data set

```
set.seed(1999)

ncells <- nrow(gene)
celltype_sim <- cell_types
celltype_sim[which(cell_types == 0)] <- 1
celltype_sim[which(cell_types == 2)] <- 2
celltype_sim[which(cell_types == 4)] <- 3
celltype_sim[which(cell_types == 6)] <- 4
condition_sim <- rep(NA, ncells)
for (ii in 4) {
  ind <- which(celltype_sim == ii)
  ind_L0 <- sample(ind, round(length(ind)*0.52), replace = F)
  ind_Y0 <- ind[!ind %in% ind_L0]
  condition_sim[ind_L0] <- "B"
  condition_sim[ind_Y0] <- "A"
}
for (ii in c(1)) {
  ind <- which(celltype_sim == ii)
  ind_L0 <- sample(ind, round(length(ind)*0.01), replace = F)
  ind_Y0 <- ind[!ind %in% ind_L0]
  condition_sim[ind_L0] <- "B"
  condition_sim[ind_Y0] <- "A"
}
for (ii in c(2:3)) {
  ind <- which(celltype_sim == ii)
  ind_L0 <- sample(ind, round(length(ind)*0.99), replace = F)
  ind_Y0 <- ind[!ind %in% ind_L0]
  condition_sim[ind_L0] <- "B"
  condition_sim[ind_Y0] <- "A"
}

celltype_sim_str <- paste0("Type ", celltype_sim)
sim_data <- pbmc_sel
sim_data@meta.data$condition_sim <- condition_sim
sim_data@meta.data$celltype_sim <- celltype_sim
sim_data@meta.data$celltype_sim_str <- celltype_sim_str
sim_data@meta.data$cell_types_str <- cell_types_str

celltype_CLE <- rep(NA, length(celltype_sim))
celltype_CLE[which(celltype_sim == 1)] <- "A"
celltype_CLE[which(celltype_sim %in% c(2,3))] <- "B"
celltype_CLE[which(celltype_sim %in% c(4))] <- "C"

sim_data@meta.data$celltype_CLE <- celltype_CLE

saveRDS(sim_data, file = paste0(data_dir, "/sim_data.rds"))
```

Visualization of the simulated data

```
count <- as.matrix(sim_data@assays$RNA@counts)
gene <- sim_data@reductions$pca@cell.embeddings
gene_umap <- sim_data@reductions$umap@cell.embeddings
condition <- sim_data@meta.data$condition_sim
cell_names <- colnames(count)
celltype_sim <- sim_data@meta.data$celltype_sim
celltype_sim_string <- sim_data@meta.data$celltype_sim_string
celltype_sim_string <- paste0("Type ", celltype_sim)

types <- paste0("Type ", 1:4)

#####

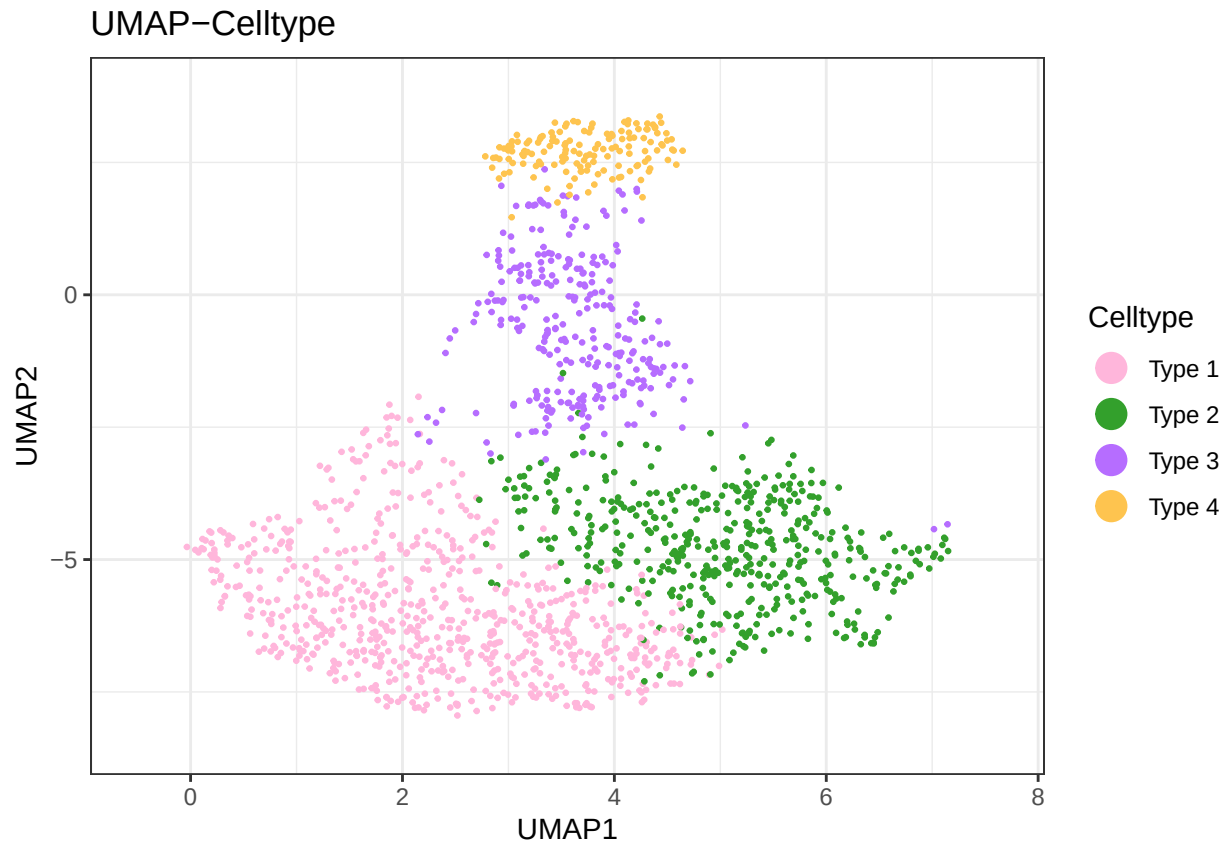
pointsize <- 0.5

df <- data.frame(PCA1 = gene[,1], PCA2 = gene[,2],
                 UMAP1 = gene_umap[,1], UMAP2 = gene_umap[,2],
                 Condition = condition,
                 Celltype = celltype_sim_string)

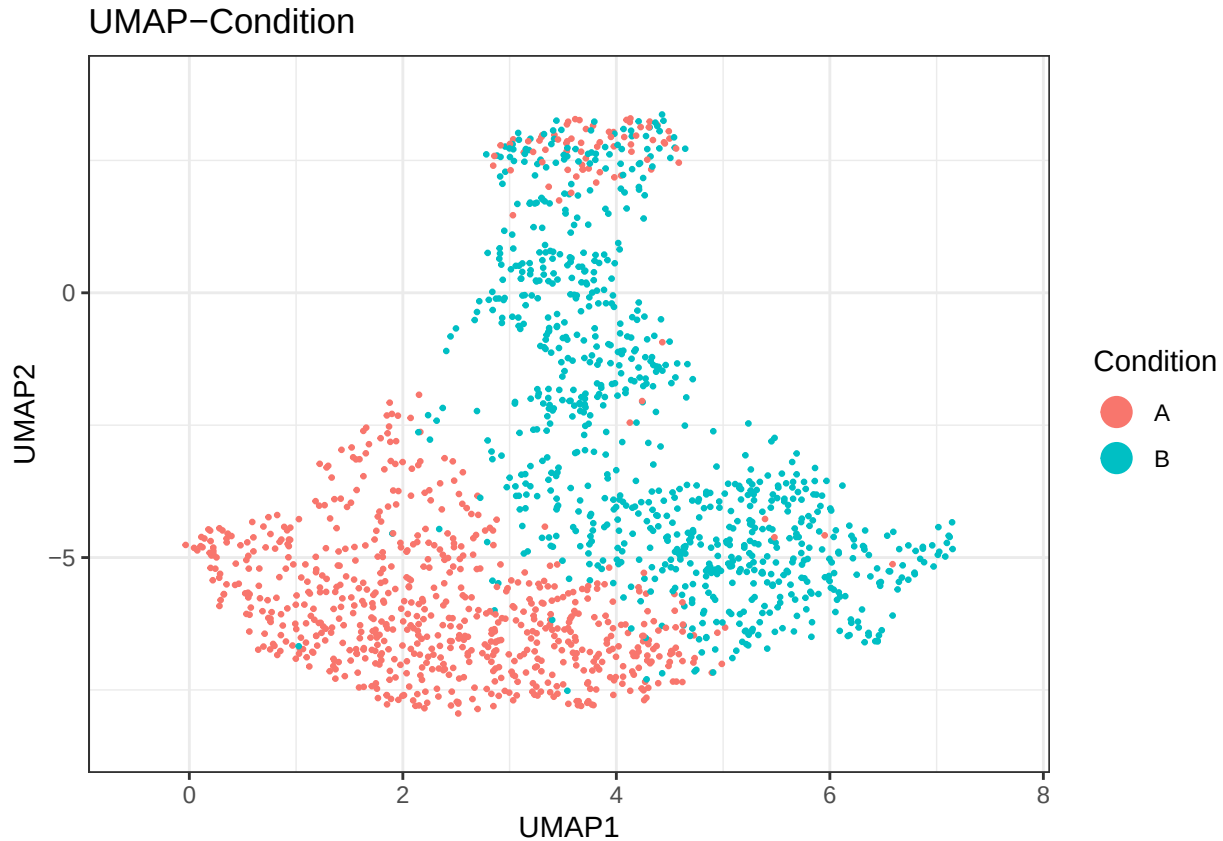
xrange <- c(min(df$UMAP1) - 0.5 , max(df$UMAP1) + 0.5)
yrange <- c(min(df$UMAP2) - 0.5 , max(df$UMAP2) + 0.5)

pal0 <- pal[1:length(types)]
names(pal0) <- types

# plot
p1 <- ggplot(df, aes(x = UMAP1, y = UMAP2, color = Celltype)) +
  geom_point(size = pointsize) +
  guides(colour=guide_legend(override.aes=list(size = 5))) +
  scale_color_manual(values=c(pal0)) +
  ylim(yrange) + xlim(xrange) +
  ggtitle("UMAP-Celltype")
p1
```



```
# plot
p2 <- ggplot(df, aes(x = UMAP1, y = UMAP2, color = Condition)) +
  geom_point(size = pointsize) +
  guides(colour=guide_legend(override.aes=list(size = 5))) +
  ylim(yrange) + xlim(xrange) +
  ggtitle("UMAP-Condition")
p2
```



```
# data

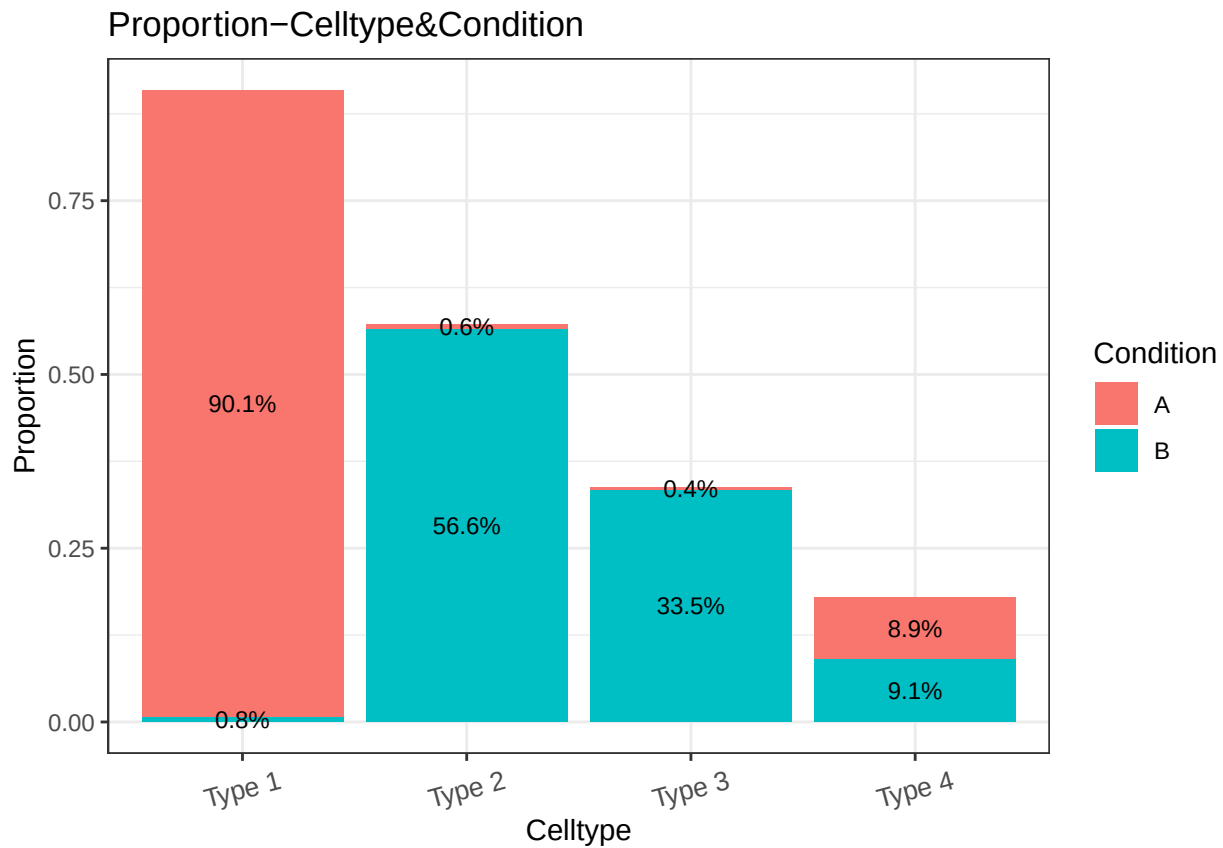
LOCRC_cells <- cell_names[which(condition == "B")]
y0 <- as.matrix(gene[LOCRC_cells, ])
YOCRC_cells <- cell_names[which(condition == "A")]
y1 <- as.matrix(gene[YOCRC_cells, ])

# prop of EO/LO in the cell types
num_types <- length(types)
num_cells_LO <- num_cells_Y0 <- num_cells <- rep(0, num_types)
names(celltype_sim_string) <- cell_names
for (ii in 1:num_types) {
  num_cells_LO[ii] <- length(which(celltype_sim_string[LOCRC_cells] == types[ii]))
  num_cells_Y0[ii] <- length(which(celltype_sim_string[YOCRC_cells] == types[ii]))
  num_cells[ii] <- length(which(celltype_sim_string == types[ii]))
}

## proportion of celltypes within condition
prop_cells <- c(num_cells_LO/sum(num_cells_LO), num_cells_Y0/sum(num_cells_Y0))
# plot
df_bar <- data.frame(Condition = c(rep("B", num_types), rep("A", num_types)),
                      Celltype = rep(types, 2),
                      Proportion = prop_cells)

cluster_names <- types
# Grouped bar plot
p3 <- ggplot(df_bar, aes(fill=Condition, y=Proportion, x=Celltype)) +
```

```
geom_bar(stat="identity") +
scale_x_discrete(limits = cluster_names) +
geom_text(aes(label=paste0(sprintf("%.1f", prop_cells*100),"%")), position = position_stack(vjust = 0.5)) +
theme(axis.text.x = element_text(angle = 15, vjust = 0.5, hjust=0.5, size = 10)) +
ggtitle("Proportion-Celltype&Condition")
p3
```

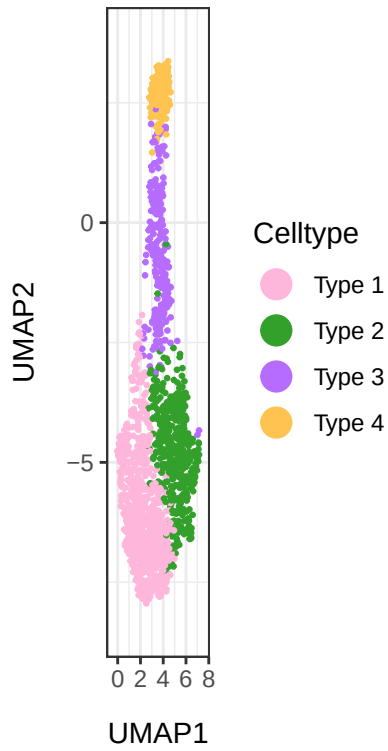


```
plot <- ggarrange(p1, p2, p3, ncol = 3, nrow = 1, align = "hv",
  labels = c("(a)", "(b)", "(c)"))

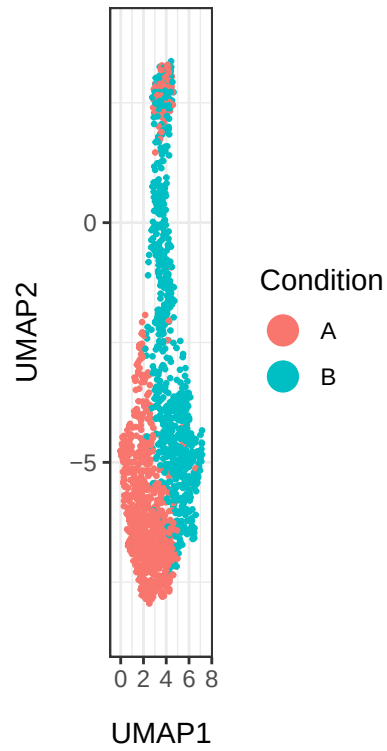
annotate_figure(plot, top = text_grob("Simulation Settings", face = "bold", size = 14))
```

Simulation Settings

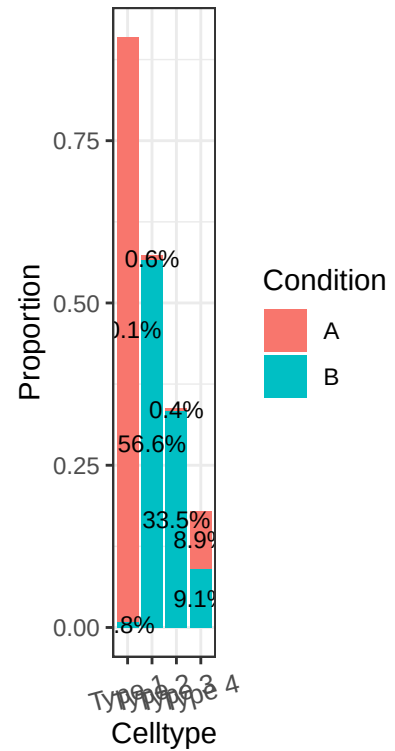
(a) UMAP-Celltype



(b) UMAP-Condition



(c) Proportion-Celltype



```
ggsave(filename = paste0(data_dir, "/figures/sim_setting.pdf"), width = 11, height = 3)
```